NMAM Institute of Technology, Nitte

Off-Campus Centre of Nitte (Deemed to be University)

COMPLETE EXAMINATION QUESTION BANK

IV Semester B.E./B.Tech (CBCS)

Academic Years: 2022–2025

Subjects:

Code	Subject
MA2005-1	Linear Algebra and Numerical Methods
CS2102-1	Database Management Systems
CS3004-1	Design and Analysis of Algorithms
CS2103-1	Software Engineering & Project Management
CS3005-1	Microprocessor and Embedded Systems
ME1654-1	Innovation and Design Thinking

Organized by: Subject → MSE-I → MSE-II → SEE

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Part I
LINEAR ALGEBRA AND NUMERICAL METHODS

1 MSE-I Examinations

1.1 MA2005-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025 Duration: 1 Hour — Max Marks: 15
MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)

1. For what values of k will the augmented matrix $\begin{bmatrix} 1 & 2 & 1 & | & 2 \\ 2 & 2 & 1 & | & 3 \\ 1 & 2 & k & | & k+1 \end{bmatrix}$ correspond to an inconsistent system of linear equations?
   (A) $k \neq 1$ (B) $k = 0$ (C) $k = 1$ (D) All values of k
2. If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, find $(AB)^{-1}$.
   (A) $\begin{bmatrix} 1 & 1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1/2 \\ -1/6 & 1/2 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -1/3 \\ 1/6 & 1/2 \end{bmatrix}$
3. Given a matrix A, column 1 and column 3 are pivot columns, then the dimension of Col A is:
   (A) 1 (B) 2 (C) 3 (D) 4

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)

UNIT – I

Q1. a) Find the eigenvalues of $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$. Also determine the eigenvectors corresponding to each eigenvalue. [6 Marks]

b) Reduce the matrix $A = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$ to the form $PA = LU$, using partial pivoting, where L is a lower triangular matrix with unit diagonal entries and U is an upper triangular matrix. [6 Marks]

UNIT – II

Q2. a) Solve by Gauss-Seidel method: $10x+y+z = 12$; $2x+10y+z = 13$; $x+y+5z = 7$. (Perform 3 iterations) [6 Marks]

b) Find the dominant eigenvalue and corresponding eigenvector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ using Rayleigh's Power method (4 iterations). [6 Marks]

1.2 MA2005-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024 Duration: 1 Hour — Max Marks: 15
MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part B: Descriptive Questions

Q1. a) Solve by Gauss-Seidel iteration method (4 iterations): $20x + y - 2z = 17$; $3x + 20y - z = -18$; $2x - 3y + 20z = 25$. Start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$.

b) Using the power method, find the dominant eigenvalue and corresponding eigenvector of $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ starting with initial value $[1, 0, 0]^T$. (4 iterations)

Q2. a) Solve the system $x + y + z = 1$; $4x + 3y - z = 6$; $3x + 5y + 3z = 4$ by LU factorization method.

b) Find all eigenvalues of $A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$. Find eigenvector for largest eigenvalue.

Q3. a) Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$.

b) Reduce the matrix $A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$ to echelon form and find its rank.

1.3 21CS401 MSE-I – June 2023

Mid Semester Examination - I, June 2023 Duration: 1 Hour — Max Marks: 20
21CS401 – LINEAR ALGEBRA

Part B: Descriptive Questions

Q1. a) Find the eigenvalues and eigenvectors of the matrix.

b) Reduce the given matrix to echelon form.

Q2. a) Solve the system of equations using Gauss elimination method.

b) Solve using LU factorization method.

Q3. a) Apply Power method to find dominant eigenvalue (4 iterations).

b) Diagonalize the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$. Hence find A^6.

2 MSE-II Examinations

2.1 MA2005-1 MSE-II – April 2025

Mid Semester Examination - II, April 2025 Duration: 1 Hour — Max Marks: 20
MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, x_1)$. Then $T^{-1}(x_1, x_2)$ equals:

   (A) $(x_2, x_1 - x_2)$ (B) $(x_2, x_1 + x_2)$ (C) $(x_1, x_1 - x_2)$ (D) $(x_1, x_1 + x_2)$

2. The set $\{1, x, x^2\}$ forms a basis for:

   (A) P_1 (B) P_2 (C) P_3 (D) P_4

3. Which of the following is a subspace of the given vector space?

   (A) The set of all polynomials of degree exactly 3, in P_3
   (B) The set of all polynomials with positive coefficients, in P
   (C) The set of all 2×2 matrices with positive entries, in $M_{2\times 2}$
   (D) The set of all symmetric 2×2 matrices, in $M_{2\times 2}$

4. Given matrix $A_{4\times 5}$ with $\text{rank}(A) = 3$, then $\dim(\text{Null}(A))$ is:

   (A) 1 (B) 2 (C) 3 (D) 4

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)

UNIT – I

Q1. a) Let $T : P_2 \to P_2$ be the transformation that maps a polynomial $p(t)$ into the polynomial $p(t+2)$. Find the image of $p(t) = 3 - 2t + t^2$. Show that T is a linear transformation. Find the matrix representation of T relative to the basis $B = \{1, t, t^2\}$. [8 Marks]

b) Let $\vec{v_1} = (1, -2, 3)$, $\vec{v_2} = (5, 6, -1)$, $\vec{v_3} = (3, 2, 1)$. Show that the set $S = \{\vec{v_1}, \vec{v_2}, \vec{v_3}\}$ is a basis for $\mathbb{R}^3$. Express each of the standard basis vectors as linear combinations of the vectors in S. [8 Marks]

UNIT – II

Q2. a) Use Gram-Schmidt process to construct orthonormal basis from the basis $B = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of $\mathbb{R}^3$. Find the QR factorization of the matrix whose column vectors form B. [8 Marks]

b) Use coordinate vectors to test the linear independence of the set of polynomials: $S = \{1 - 2t^2 - t^3, t + 2t^3, 1 + t - 2t^2\}$. [8 Marks]

2.2 21CS401 MSE-II – July 2023

Mid Semester Examination - II, July 2023 Duration: 1 Hour — Max Marks: 20
21CS401 – LINEAR ALGEBRA

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. The dimension of the vector space of all symmetric $n \times n$ matrices is:

   (A) n^2 (B) $\frac{n(n+1)}{2}$ (C) $\frac{n(n-1)}{2}$ (D) n

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x + y, x - y)$. Then T is:

   (A) Linear but not one-to-one (B) One-to-one but not onto
   (C) Both one-to-one and onto (D) Neither one-to-one nor onto

3. If $\{v_1, v_2, v_3\}$ is a basis for V, then $\{v_1 + v_2, v_2 + v_3, v_3 + v_1\}$ is:

   (A) Always a basis (B) Never a basis (C) A basis if vectors are orthogonal (D) Cannot be determined

4. The null space of an $m \times n$ matrix A is a subspace of:

   (A) $\mathbb{R}^m$ (B) $\mathbb{R}^n$ (C) $\mathbb{R}^{m+n}$ (D) $\mathbb{R}^{mn}$

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)

Q1. a) Let $V = P_2$ (polynomials of degree ≤ 2). Show that $S = \{1 + x, x + x^2, 1 + x^2\}$ is a basis for V. [8 Marks]

b) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x + 2y - z, 2x - y + z)$. Find the matrix of T with respect to standard bases. Find $\ker(T)$ and $\text{range}(T)$. [8 Marks]

Q2. a) Use Gram-Schmidt process to orthonormalize the vectors $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$. [8 Marks]

b) Find the least squares solution of $A\vec{x} = \vec{b}$ where $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$. [8 Marks]

3 SEE (Semester End Examinations)

3.1 MA2005-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part A: Multiple Choice Questions ($20 \times 1 = 20$ Marks)

1. For what value of k, the augmented matrix $\begin{bmatrix} 1 & 3 & -2 & | & 3 \\ 3 & k & -6 & | & 9 \\ 1 & -3 & 2 & | & 5 \end{bmatrix}$ corresponds to an inconsistent system?

   (A) $k = 1$ (B) $k = 9$ (C) $k = -3$ (D) None of these

2. If A and B are two matrices such that AB exists and is a square matrix, then:

   (A) A and B are square matrices of same order
   (B) Either both are square or number of rows of A = number of columns of B
   (C) BA need not exist
   (D) Any one of A and B is identity

3. Columns 1, 2 and 5 are pivot columns of a matrix A. Which is NOT necessarily true?

   (A) $\dim(\text{Col } A) = 3$ (B) $\dim(\text{Row } A) = 3$ (C) $\text{rank}(A) = 3$ (D) $\dim(\text{Null } A) = 3$

4. If λ is an eigenvalue of A, then eigenvalue of A^{-1} is:

   (A) λ (B) $-\lambda$ (C) λ^2 (D) $1/\lambda$

5. If $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ then $A^3 - 3A^2 - A + 3I =$

   (A) $A + I$ (B) I (C) $2A$ (D) 0

6. The eigenvalues of $P^{-1}AP$ are:

   (A) Same as that of PA (B) Same as that of A (C) Same as that of P (D) Cannot say

7. Correct order to solve by Gauss-Seidel method:

   (A) Arrange in any order (B) Diagonal elements are zero
   (C) Diagonal elements larger than other elements (D) Cannot be solved

8. Which is a direct method for solving system of linear equations?

   (A) Jacobi (B) Gauss-Seidel (C) LU Decomposition (D) Power method

9. If λ is eigenvalue of A with eigenvector $\vec{x}$, then $\lambda^2 + 1$ is eigenvalue of:

   (A) $A^2 + I$ (B) $A^2 + A$ (C) $2A$ (D) $(A + I)^2$

10. If $1 + x$, $1 + x + x^2$, and $ax^2 + x^3$ are linearly dependent, then $a =$:

    (A) 1 (B) 2 (C) 3 (D) 4

11. The dimension of subspace $W = \{(a, b, c, d) : a - 2b + c - d = 0\}$ is:

    (A) 1 (B) 2 (C) 3 (D) 4

12. Let V be set of all 2×2 symmetric matrices. Then dimension of V is:

    (A) 1 (B) 2 (C) 3 (D) 4

13. For linear transformation $T : V \to W$, let U be subspace of W. Then $T^{-1}(U)$ is:

    (A) Subspace of V (B) Subspace of W (C) Subspace of both (D) Not a subspace

14. Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented as matrix multiplication.

    (A) True only if $m = n$ (B) True only if $m \neq n$ (C) Always true (D) Never true

15. If $\vec{u}$ and $\vec{v}$ are orthogonal vectors in $\mathbb{R}^n$, then:

    (A) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$ (B) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 - \|\vec{v}\|^2$
    (C) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 \cdot \|\vec{v}\|^2$ (D) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 / \|\vec{v}\|^2$

16. Two vectors $\vec{u} = (1, 2, -3)$ and $\vec{v} = (a, 1, 1)$ are orthogonal if $a =$:

    (A) 1 (B) 2 (C) 3 (D) 5

17. A linear combination of the rows of A can be written as:

    (A) $\vec{x}^T A$ (B) $A\vec{x}$ (C) $\vec{x}A^T$ (D) $A^T\vec{x}$

18. The angle between vectors $\vec{u} = (1, 2, -1, 0)$ and $\vec{v} = (-1, 1, 1, 2)$ is:

    (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) $2\pi/3$

19. Which defines an inner product on $\mathbb{R}^2$?

    (A) $\langle \vec{u}, \vec{v} \rangle = u_1 v_1 - u_2 v_2$ (B) $\langle \vec{u}, \vec{v} \rangle = u_1 v_1 + 2u_2 v_2$
    (C) $\langle \vec{u}, \vec{v} \rangle = u_1^2 v_1^2 + u_2^2 v_2^2$ (D) $\langle \vec{u}, \vec{v} \rangle = u_1 v_2 - u_2 v_1$

20. For orthonormal basis $\{\vec{u}_1, \ldots, \vec{u}_n\}$ of $\mathbb{R}^n$ and any $\vec{x} \in \mathbb{R}^n$:

    (A) $\|\vec{x}\|^2 = \sum_{i=1}^n |\langle \vec{x}, \vec{u}_i \rangle|^2$ (B) $\|\vec{x}\|^2 = \sum_{i=1}^n \langle \vec{x}, \vec{u}_i \rangle$
    (C) $\|\vec{x}\| = \sum_{i=1}^n |\langle \vec{x}, \vec{u}_i \rangle|$ (D) None of the above

Part B: Descriptive Questions ($5 \times 16 = 80$ Marks)

UNIT – I

Q1. a) Find basis for Nul A and Col A where $A = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 3 \\ 1 & 3 & 1 & 1 & 2 \\ -2 & -6 & 0 & -2 & 4 \end{bmatrix}$ [8 Marks]

b) Find eigenvalues of $A = \begin{bmatrix} 0 & -4 & -6 \\ -1 & 0 & -3 \\ 1 & 2 & 5 \end{bmatrix}$. Find eigenvectors and verify Cayley-Hamilton theorem. [8 Marks]

Q2. a) Compute A^{10} if $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. [8 Marks]

b) Reduce $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ to $A = LU$. Use factorization to solve $A\vec{x} = \vec{b}$ where $\vec{b} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. [8 Marks]

UNIT – II

Q3. a) Solve by Gauss-Seidel method (4 iterations): $10x - y + 2z = 6$; $-x + 11y - z + 3w = 25$; $2x - y + 10z - w = -11$; $3y - z + 8w = 15$. [8 Marks]

b) Find dominant eigenvalue and eigenvector of $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using Power method (5 iterations). [8 Marks]

Q4. a) Let $V = M_{2\times 2}$. Show $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a + d = 0 \right\}$ is subspace of V. Find its dimension. [8 Marks]

b) Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be $T(x_1, x_2) = (x_1 - x_2, x_1, 2x_1 + x_2)$. Show T is linear. Find matrix. Is T one-to-one? Is T onto? [8 Marks]

UNIT – III

Q5. a) Find QR factorization of $A = \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$. [8 Marks]

b) For $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, find the SVD of A. [8 Marks]

Q6. a) Find best line $y = \beta_0 + \beta_1 x$ for data points: $(0, 1), (1, 1), (2, 2), (3, 2)$ using least squares. [8 Marks]

b) Let $S = \{(2, 5, -3, -1), (-2, -3, 2, -5), (1, 3, -2, 0), (-1, -6, 3, 3)\}$. Find basis for subspace of $\mathbb{R}^4$ spanned by S. [8 Marks]

Part II
DATABASE MANAGEMENT SYSTEMS

4 MSE-I Examinations

4.1 CS2102-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025 Duration: 1 Hour — Max Marks: 15
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part A: Multiple Choice Questions (3 × 1 = 3 Marks)

1. A _____ is a unit of storage containing one or more records.
   (A) Block (B) Field (C) Tuple (D) File
2. Which of the following is NOT a component of ER Model?
   (A) Entity (B) Attribute (C) Table (D) Relationship
3. Which of the following is used to uniquely identify a record in a table?
   (A) Foreign Key (B) Candidate Key (C) Primary Key (D) Super Key

Part B: Descriptive Questions (2 × 6 = 12 Marks)

UNIT – I

Q1. a) Design and construct an ER diagram for "Student Database Management System". Indicate all keys, constraints and assumptions. [6 Marks]

b) Discuss the main characteristics of the database approach. [6 Marks]

Q2. a) Identify the Entity, Relationship, Participation Constraint and Cardinality ratio for the ER diagram of Movie database schema (ACTOR, MOVIE, DIRECTOR, PRODUCER entities with relationships). [6 Marks]

b) Describe Three Schema Architecture with a neat diagram. [6 Marks]

UNIT – II

Q3. a) Write the following queries in Relational Algebra for:

```
EMPLOYEE(SSN, SNAME, AGE, SALARY, DEPT_ID)
DEPARTMENT(DEPT_ID, DEPT_NAME, DEPT_HEAD)
PROJECT(PID, DEPT_ID, PNAME)
WORKSON(SSN, PID, PHOURS)
```

(i) Get names of employees working on projects managed by 'HR' department.
(ii) Retrieve names of employees who do not work on any project.
(iii) Find names of all departments that have at least one project. [6 Marks]

b) Explain how to deal with constraints violation during insertion and delete operations. [6 Marks]

4.2 CS2102-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024 Duration: 1 Hour — Max Marks: 15
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part B: Descriptive Questions

Q1. a) Describe Three Schema Architecture with diagram. Why do we need mappings among schema levels? [6 Marks]

b) Design ER diagram for "Engineering College Management System". [6 Marks]

Q2. a) Discuss the main characteristics of database approach. [6 Marks]

b) Explain domain constraints and referential integrity constraints in relational model. [6 Marks]

Q3. a) Define Entity attribute. Explain different types of attributes in ER diagram. [6 Marks]

b) Outline steps to convert basic ER model to relational database scheme. [6 Marks]

Q4. a) "For good database design, matching attributes must be primary key or foreign key otherwise it leads to spurious tuples". Justify. [6 Marks]

b) Discuss the characteristics of relations. [6 Marks]

5 MSE-II Examinations

5.1 CS2102-1 MSE-II – April 2025

Mid Semester Examination - II, April 2025 Duration: 1 Hour — Max Marks: 20
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. The rule "No component of Primary Key accepts Null value" is:

   (A) Referential Integrity Constraint (B) Domain Integrity Constraint
   (C) Entity Integrity Constraint (D) Tuple Integrity Constraint

2. Which command adds a new column to existing table?

   (A) ALTER (B) UPDATE (C) INSERT (D) MODIFY

3. A relation is in BCNF if every non-trivial FD $X \to Y$:

   (A) X is a super key (B) Y is a prime attribute
   (C) X is a candidate key (D) Y is a super key

4. The operation to combine tuples from two relations based on condition:

   (A) Union (B) Join (C) Projection (D) Selection

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)

Q1. Consider the schema and write SQL queries:

```
Suppliers(sid: integer, sname: string, address: string)
Parts(pid: integer, pname: string, color: string)
Catalog(sid: integer, pid: integer, cost: double)
```

(i) Find the sids of suppliers who supply some red or green part.
(ii) Find the sids of suppliers who supply only red parts.
(iii) Find the pnames of parts for which there is some supplier.
(iv) Find the sids of suppliers who supply some red part or are at 221 Packer Street. (Nested Query) [8 Marks]

Q2. Consider the schema and write SQL queries:

```
Movie(movie_id, title, release_year, genre, director_id, rating)
Director(director_id, name, birth_year, nationality)
Actor(actor_id, name, birth_year, nationality)
Movie_Actor(movie_id, actor_id, role)
Review(rev_id, user_id, movie_id, rev_text, rating, rev_date)
User(user_id, username, email, age)
```

(i) Find names of actors who acted in at least one movie released after 2015.
(ii) Find directors who directed more than 5 movies.
(iii) Get average rating of movies directed by 'Christopher Nolan'.
(iv) Find movies with higher rating than average rating of same genre. (Nested Query) [8 Marks]

Q3. a) Consider FDs $F = \{A \to C, AC \to D, E \to AD, E \to H\}$ and $G = \{A \to CD, E \to AH\}$. Are they equivalent? [8 Marks]

b) What is Normalization? Explain First Normal Form with example. [8 Marks]

Q4. a) Given $R(ABCDEFGHIJ)$ and FDs: $\{(A, B) \to C, A \to \{D, E\}, B \to F, F \to \{G, H\}, D \to \{I, J\}\}$. Find candidate key. Check if relation is in 3NF. If not, decompose to 3NF. [8 Marks]

b) Explain any two Informal Design Guidelines for Relation Schemas. [8 Marks]

6 SEE (Semester End Examinations)

6.1 CS2102-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part A: Multiple Choice Questions (20 × 1 = 20 Marks)

1. Which is not a characteristic of DBMS? (A) Data Independence (B) Concurrent Access (C) Data Redundancy (D) Data Security
2. Weak entity is represented by: (A) Rectangle (B) Double Rectangle (C) Diamond (D) Oval
3. Grouping related objects in ER model: (A) Aggregation (B) Generalization (C) Specialization (D) Composition
4. Each row of table represents a: (A) Field (B) Record (C) Attribute (D) Key
5. Retrieve data from multiple tables: (A) Union (B) Intersection (C) Cartesian Product (D) Projection
6. Produces relation with only specified attributes: (A) Selection (B) Projection (C) Join (D) Division
7. Filter groups formed by GROUP BY: (A) WHERE (B) ORDER BY (C) HAVING (D) FILTER
8. NOT a DML command: (A) SELECT (B) INSERT (C) CREATE (D) UPDATE
9. Trivial FD $X \to Y$ if: (A) $X \subseteq Y$ (B) $Y \subseteq X$ (C) $X = Y$ (D) $X \cap Y = \phi$
10. Eliminates transitive dependencies: (A) 1NF (B) 2NF (C) 3NF (D) BCNF
11. Closure of attributes X under FDs F: (A) X^+ (B) F^+ (C) X^* (D) F^*
12. 2NF requires 1NF and: (A) No partial dependencies (B) No transitive dependencies (C) Every determinant is candidate key (D) All attributes atomic

13. Lossless join decomposition uses: (A) Union (B) Intersection (C) Natural join (D) Cartesian product
14. NOT property of good decomposition: (A) Lossless join (B) Dependency preservation (C) Redundancy (D) No anomalies
15. Minimal cover has: (A) Single attribute on RHS (B) No redundant FDs (C) No extraneous attributes on LHS (D) All of the above
16. Permanently save transaction changes: (A) SAVEPOINT (B) COMMIT (C) ROLLBACK (D) GRANT
17. ACID properties include: (A) Atomicity, Consistency, Isolation, Durability (B) Atomicity, Concurrency, Isolation, Durability (C) Accuracy, Consistency, Integrity, Durability (D) Atomicity, Consistency, Integrity, Durability
18. Conflict serializable if: (A) Can transform to serial schedule by swapping non-conflicting operations (B) All transactions execute serially (C) No conflicts (D) Precedence graph has no cycles
19. Guarantees deadlock freedom: (A) Simple Lock (B) Two-Phase Locking (C) Strict 2PL (D) Wait-Die Scheme
20. Timestamp ordering for write on X: (A) Always allowed (B) If $TS > W$-timestamp(X) and $TS > R$-timestamp(X) (C) If $TS < W$-timestamp(X) (D) Never allowed

Part B: Descriptive Questions (80 Marks)

Q1. a) What are Characteristics of Database approach? Explain. [8 Marks]
b) With neat diagram, explain Three Schema architecture. [8 Marks]
Q2. a) Explain attribute types: (i) Composite (ii) Multivalued (iii) Atomic. [8 Marks]
b) Explain Cardinality Ratio for Binary relationships with examples. [8 Marks]
Q3. a) Consider STUDENT and COURSE tables. Write Relational Algebra queries. [8 Marks]
b) Explain steps in ER to Relational Mapping Algorithm. [8 Marks]
Q4. a) Explain Not Null, DEFAULT and CHECK constraints in SQL. [8 Marks]
b) Explain CASCADE and RESTRICT options of DROP TABLE. [8 Marks]
Q5. a) What are Informal Design Guidelines for relational schema? [8 Marks]
b) What is Functional Dependency? Explain with example. [8 Marks]
Q6. a) Explain Standard Inference rules for Functional Dependencies. [8 Marks]
b) Explain algorithm for determining X^+, Closure of X under F. [8 Marks]
Q7. a) What is B+ Tree? Explain structure & characteristics with diagram. [8 Marks]
b) Briefly explain ACID properties. [8 Marks]
Q8. a) What is Query Optimizer? Explain working with diagram. [8 Marks]
b) What are 2 rules of Strict 2PL? Explain. [8 Marks]

6.2 CS2102-1 SEE – July 2024

Semester End Examination, July 2024 Duration: 3 Hours — Max Marks: 100
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part B: Descriptive Questions

Q1. a) Briefly explain various characteristics of database approach. [8 Marks]

b) Explain Relational Algebra operations: (i) Division (ii) Union (iii) Cartesian product (iv) COUNT. [8 Marks]

Q2. a) Design ER diagram and schema for flight management system. [8 Marks]

b) Explain grouping concepts in relational algebra. [8 Marks]

Q3. a) With block diagram explain three schema architecture. [8 Marks]

b) Sailors-Boats-Reserves queries in Relational Algebra. [8 Marks]

Q4. a) Employee-Works-Company-Manages SQL queries. [8 Marks]

b) What is a view? Give specification in SQL. [8 Marks]

Q5. a) Check lossless join property for given decompositions. [8 Marks]

b) How basic constraints are specified in SQL-99. [8 Marks]

Q6. a) Order processing database SQL queries. [8 Marks]

b) Need for normalization. Explain 1NF, 2NF, 3NF with examples. [8 Marks]

Q7. a) Explain Strict Two-Phase Locking protocol. [8 Marks]

b) Explain index structures: Hash based, Tree based indexing. [8 Marks]

Q8. a) Illustrate anomalies with Interleaved Execution. [8 Marks]

b) Algorithm to insert element in B+ tree. [8 Marks]

Part III
DESIGN AND ANALYSIS OF ALGORITHMS

7 MSE-I Examinations

7.1 CS3004-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025 Duration: 1 Hour — Max Marks: 15
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)

1. If algorithm has $T(n) = 7n^3 + 10n^2 + 4$, then $T(n)$ belongs to:
   (A) $O(n)$ (B) $O(n^2)$ (C) $O(n^3)$ (D) $O(n^4)$
2. Time complexity of $T(n) = 2T(n/2) + n$:
   (A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $O(\log n)$
3. Minimum comparisons to find minimum in array of n elements:
   (A) n (B) $n - 1$ (C) $n + 1$ (D) $2n$

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)

UNIT – I

Q1. a) Summarize algorithm design and analysis process with diagram. [6 Marks]
b) For given algorithm, determine: (i) What it computes (ii) Basic operation (iii) How many times executed (iv) Efficiency. [6 Marks]
Q2. a) Define asymptotic notations Big Oh (O) and Big Theta (Θ) with examples. [6 Marks]
b) Show that efficiency of Tower of Hanoi is exponential. [6 Marks]
UNIT – II
Q3. a) Develop brute force string matching algorithm. Analyze worst-case complexity. [6 Marks]
b) Outline merge sort algorithm and derive time complexity. [6 Marks]
Q4. a) Define TSP. Solve using exhaustive search for 4-vertex graph with edges: A-B=6, A-C=3, A-D=1, B-C=4, B-D=2, C-D=8. State efficiency. [6 Marks]
b) Apply Quick sort on marks: 8, 10, 5, 2, 14. Show tree of recursive calls. [6 Marks]

7.2 CS3004-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024 Duration: 1 Hour — Max Marks: 15
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part B: Descriptive Questions

Q1. a) Explain properties of algorithm using linear search example. [6 Marks]
b) For each function, indicate change when argument increased fourfold: (i) $\log_2 n$ (ii) n (iii) n^2 (iv) 2^n (v) $\sqrt{n}$. [6 Marks]

Q2. a) Explain general framework for Analysis of an algorithm. [6 Marks]
b) Define 3 asymptotic notations. Check: (i) $10n^2 + 9 \in O(n^2)$ (ii) $n(n+1)/2 \in O(n^2)$. [6 Marks]
Q3. a) Write general plan for analyzing recursive algorithm. Find factorial efficiency. [6 Marks]
b) Give two brute force algorithms to sort integers. [6 Marks]
Q4. a) Illustrate Merge sort and analyze efficiency. Sort: 6, 14, 8, 2, 15, 10, 7, 5. [6 Marks]
b) Show comparisons for brute force string matching: P=0001 in T=000010001010101. [6 Marks]

7.3 21CS402 MSE-I – June 2023

Mid Semester Examination - I, June 2023 Duration: 1 Hour — Max Marks: 20
21CS402 – DESIGN AND ANALYSIS OF ALGORITHMS

Part B: Descriptive Questions

Q1. a) With process flow diagram explain algorithm design and analysis. [8 Marks]
b) Design recursive algorithm for $2^n = 2^{n-1} + 2^{n-1}$. Find complexity. [8 Marks]
Q2. a) Define three asymptotic notations. Express $f(n) = n(n-1)/2$ using three notations with proof. [8 Marks]
b) Perform complexity analysis of sequential search for three efficiency types. [8 Marks]
Q3. a) What is brute force? Give brute force string matching with complexity. Trace: TEXT: JIM SAW ME IN SHOE SHOP, PATTERN: SHE. [8 Marks]
b) Apply merge sort to L,A,S,T,S,O,R,T,I,N,G alphabetically. Analyze complexity. [8 Marks]
Q4. a) Give complete Quick sort algorithm with complexity analysis. Trace for: 5, 3, 1, 9, 8, 2, 4, 7. [8 Marks]
b) What is exhaustive search? Solve knapsack: $n = 4$, capacity $m = 10$, $w = (4, 6, 3, 5)$, values $= (30, 50, 20, 10)$. [8 Marks]

8 MSE-II Examinations

8.1 CS3004-1 MSE-II – April 2025

Mid Semester Examination - II, April 2025 Duration: 1 Hour — Max Marks: 20
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. Necessary condition for topological sorting:
   (A) Graph must be tree (B) Graph must be DAG
   (C) Graph must be complete (D) Graph must be bipartite
2. Data structure used in DFS:
   (A) Queue (B) Heap (C) Priority queue (D) Stack
3. Time complexity of distribution counting sort:
   (A) $\Theta(n^2)$ (B) $O(n \log n)$ (C) $O(2^n)$ (D) $\Theta(n)$

4. Dynamic programming builds solution from:

   (A) Overlapping subproblems (B) Single recursive call
   (C) Greedy approach (D) Divide and conquer

Part B: Descriptive Questions (2 × 8 = 16 Marks)

Q1. a) Outline Insertion sort algorithm. Show sorting of E, X, A, M, P, L, E in alphabetical order. [4 Marks]

b) Apply heap sort to: 90, 89, 56, 72, 67, 12, 21, 34, 45. [4 Marks]

Q2. a) Define AVL tree. Construct AVL tree for $\{2, 9, 3, 12, 15, 8, 11\}$. [4 Marks]

b) Define topological sorting. Illustrate using Source Removal method for given graph. [4 Marks]

Q3. a) Write Comparison Counting sort algorithm. Compare complexity with Selection and Bubble sort. [4 Marks]

b) Summarize Horspool algorithm. Search TIME in "A STITCH IN TIME SAVES NINE". [4 Marks]

Q4. a) Apply All Pair Shortest Path algorithm for given weighted graph. [4 Marks]

b) Write BFS algorithm. List two facts differentiating BFS from DFS. [4 Marks]

9 SEE (Semester End Examinations)

9.1 CS3004-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions (20 × 1 = 20 Marks)

1. Quick sort divides array into: (A) One (B) Two (C) Three (D) Four
2. Binary Search works on: (A) Unsorted (B) Sorted (C) Distinct elements (D) Integers only
3. Linear search is also called: (A) Binary search (B) Simple search (C) Sequential search (D) Difficult search
4. $O(n)$ means: (A) Exponential (B) Linear growth (C) Independent of input (D) Logarithmic
5. Output of fun(5) where fun(n) returns 1 if n==1, else n+fun(n-1): (A) 120 (B) 24 (C) 6 (D) 15
6. Pseudocode uses: (A) Native + programming language (B) High level only (C) Natural language only (D) Low level
7. NOT a characteristic of good algorithm: (A) Clarity (B) Efficiency (C) Correctness (D) Complex data structures
8. Primary characteristic of Divide and Conquer: (A) Iteration (B) Overlapping subproblems (C) Dividing into independent subproblems, solving recursively, combining (D) Linear solving
9. 0/1 Knapsack uses: (A) Greedy (B) Divide and conquer (C) Dynamic programming (D) DFS
10. Floyd-Warshall finds: (A) Shortest path between two nodes (B) Shortest paths between all pairs (C) Longest path (D) Pattern matching
11. Horspool uses which table: (A) Shift Table (B) Hash Table (C) Match Table (D) Character Table

12. Counting Sort is distribution-based because: (A) Uses frequencies for positions (B) Divides into subarrays (C) Compares pairs (D) Recursively splits
13. Max-heap root from [3, 9, 2, 1, 4, 8]: (A) 1 (B) 9 (C) 2 (D) 8
14. AVL rotation when balance factor > 1 or < -1: (A) Rotation (B) Deletion (C) Insertion (D) Search
15. Binary Search is: (A) Decrease by constant (B) Decrease by constant factor (C) Decrease by one (D) Divide and conquer
16. In max-heap: (A) Parent smaller (B) Parent greater or equal (C) BST (D) Ascending order
17. Job Assignment minimizes: (A) Time (B) Number of tasks (C) Total cost (D) Total time worked
18. Subset-sum backtracks when: (A) Sum exceeds target (B) Sum equals target (C) All elements considered (D) Subset found
19. Dijkstra's Algorithm is: (A) Greedy (B) Branch and bound (C) Backtracking (D) Dynamic programming
20. Huffman codes use: (A) Exhaustive search (B) Greedy (C) Brute force (D) Divide and conquer

Part B: Descriptive Questions (80 Marks)

UNIT – I

Q1. a) Explain Performance Analysis of algorithm. [8 Marks]

b) Write Selection sort pseudocode and analyze. Sort: 5, 4, 2, 1, 6, 3. [8 Marks]

Q2. a) General plan for non-recursive algorithm analysis. Find maximal element and analyze. [8 Marks]

b) Write Merge sort algorithm. Derive time complexity using recurrence. Sort: 38, 27, 43, 3, 9, 82, 10. [8 Marks]

UNIT – II

Q3. a) Write insertion sort. Construct heap for: 63, 9, 19, 7, 90, 81, 45. [8 Marks]

b) Identify "JIM_SAM" using Horspool in "JIM_SAW_ME_IN_A_BARBERSHOP". [8 Marks]

Q4. a) Write B-Tree insertion algorithm. Draw B-Tree of order 5 for: 3, 14, 7, 1, 8, 5, 11, 17, 13, 6, 23, 12, 20, 26, 4, 16, 18, 24, 25, 19. [8 Marks]

b) Define Topological Sorting. Illustrate using source removal method. [8 Marks]

UNIT – III

Q5. a) Write Divide and Conquer algorithm for key comparisons. Derive complexity. Quick sort: 38, 27, 43, 3, 9, 82, 10. [8 Marks]

b) Define Class P, Class NP, NP-Complete, NP-Hard. [8 Marks]

Q6. a) Solve Job assignment using Branch and Bound for matrix: $\begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix}$ [8 Marks]

b) Solve subset sum using backtracking: $S = \{5, 10, 12, 13, 15, 18\}$, $d = 30$. [8 Marks]

Part IV
SOFTWARE ENGINEERING & PROJECT MANAGEMENT

10 MSE-I Examinations

10.1 CS2103-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025 Duration: 1 Hour — Max Marks: 15
CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

Q1. a) List and explain attributes of a good software. [6 Marks]
b) Explain Incremental Development with diagram. [6 Marks]
Q2. a) Outline ACM/IEEE Code of Ethics. [6 Marks]
b) Explain Boehm's Spiral Model with diagram. [6 Marks]
Q3. a) Outline Structure of Requirements Document. [6 Marks]
b) With diagram explain types of Non-Functional Requirements. [6 Marks]
Q4. a) Explain Requirements Elicitation and Analysis Process with diagram. [6 Marks]
b) Develop Scenario for collecting Medical history in MHC-PMS. Draw Use cases. [6 Marks]

10.2 CS2103-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024 Duration: 1 Hour — Max Marks: 15
CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

Q1. a) Discuss ethical issues in professional environment. Explain types of software applications. [6 Marks]
b) Explain spiral software process model with diagram. [6 Marks]
Q2. a) Mention essential attributes of good software. Explain SRS document structure. [6 Marks]
b) Explain incremental process model. Differences between waterfall and incremental. [6 Marks]
Q3. a) Define requirement. Discuss types of non-functional requirements with examples. [6 Marks]
b) Explain need of software engineering discipline with justifications. [6 Marks]
Q4. a) List readers of system requirements. Explain user and system requirements for MHC-PMS. [6 Marks]
b) Discuss requirement elicitation and analysis activity in RE process. [6 Marks]

10.3 21CS404 MSE-I – June 2023

Mid Semester Examination - I, June 2023 Duration: 1 Hour — Max Marks: 20
21CS404 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

Q1. a) Analyze essential attributes of good software. [8 Marks]
b) Explain various types of non-functional requirements. [8 Marks]
Q2. a) Discuss Waterfall model and give two advantages. [8 Marks]
b) Explain different types of applications that may be developed. [8 Marks]
Q3. a) Analyze areas where standards not bound by laws but by professional responsibility. [8 Marks]
b) Illustrate Boehm's spiral model. [8 Marks]
Q4. a) Build structure of requirements document. [8 Marks]
b) Show process model of elicitation and analysis process. [8 Marks]

11 MSE-II Examinations

[MSE-II papers for SEPM would go here if available in the source materials]

12 SEE (Semester End Examinations)

12.1 CS2103-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

Q1. a) Briefly explain essential attributes of good software. List fundamental SE activities. [8 Marks]
b) Explain Incremental Development with diagram. [8 Marks]
Q2. a) Outline ACM/IEEE Code of Ethics. [8 Marks]
b) Explain Boehm's Spiral Model with diagram. [8 Marks]
Q3. a) Outline Structure of Requirements Document. [8 Marks]
b) With diagram explain types of Non-Functional Requirements. [8 Marks]
Q4. a) Explain Requirements Elicitation and Analysis Process with diagram. [8 Marks]
b) Develop Scenario for MHC-PMS medical history. Draw Use cases. [8 Marks]
Q5. a) Briefly explain System Testing. [8 Marks]
b) Describe five key UML diagram types with purposes. [8 Marks]
Q6. a) Briefly explain concept of system models. [8 Marks]
b) Explain language processing system architecture. [8 Marks]
Q7. a) Briefly explain four test strategies for software development. [8 Marks]
b) Explain ATM System software architecture in Transaction Processing Systems. [8 Marks]
Q8. a) Explain project scheduling concept and significance. [8 Marks]
b) Explain configuration management concept. [8 Marks]
Q9. a) List and explain key stages in component measurement process. [8 Marks]
b) Describe various risk identification in project management. [8 Marks]

12.2 CS2103-1 SEE – May 2024

Semester End Examination, May 2024 Duration: 3 Hours — Max Marks: 100
CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

Q1. a) Explain software development life cycle. [8 Marks]
b) Differentiate Waterfall model and Spiral model. [8 Marks]
Q2. a) Explain V-model in plan-driven software process. [8 Marks]
b) Explain Boehm's spiral model. [8 Marks]
Q3. a) Distinguish functional and non-functional requirements. Explain types with diagram. [8 Marks]
b) Explain incremental model. Advantages and disadvantages. [8 Marks]
Q4. a) Explain data flow model with insulin pump example. [8 Marks]
b) Explain acceptance testing process stages with diagram. [8 Marks]
Q5. a) Explain design models: (i) Sequence models (ii) Use-case models. [8 Marks]
b) Explain extreme programming and practices. [8 Marks]
Q6. a) Identify Weather Station object classes. Draw class diagrams. [8 Marks]
b) What are agile methods? Describe principles. [8 Marks]
Q7. a) What is Risk and Risk management? Explain categories. [8 Marks]
b) Explain project planning process with diagram. [8 Marks]
Q8. a) Explain phases of software review process with diagram. [8 Marks]
b) Explain software product metrics and types. Describe static metrics. [8 Marks]

Part V
MICROPROCESSOR AND EMBEDDED SYSTEMS

13 MSE-I Examinations

13.1 CS3005-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024 Duration: 1 Hour — Max Marks: 15
CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

Q1. a) Explain internal architecture of 8086 microprocessor with diagram. [6 Marks]
b) Explain different addressing modes of 8086 with examples. [6 Marks]
Q2. a) Explain memory segmentation in 8086. [6 Marks]
b) Write ALP to find sum of array elements. [6 Marks]

13.2 21CS403 MSE-I – June 2023

Mid Semester Examination - I, June 2023 Duration: 1 Hour — Max Marks: 20
21CS403 – MICROPROCESSOR AND MICROCONTROLLERS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. The 8086 microprocessor has ___ bit data bus: (A) 8 (B) 16 (C) 20 (D) 32
2. Which register is used as loop counter in 8086: (A) AX (B) BX (C) CX (D) DX
3. Physical address in 8086 is calculated as: (A) Segment $\times$ 16 + Offset (B) Segment + Offset $\times$ 16 (C) Segment + Offset (D) Segment $\times$ Offset
4. Flag set when result is zero: (A) Carry Flag (B) Zero Flag (C) Sign Flag (D) Overflow Flag

Part B: Descriptive Questions

Q1. a) Explain internal architecture of 8086 with diagram. [8 Marks]
b) Explain different addressing modes of 8086 with examples. [8 Marks]
Q2. a) Write 8086 ALP to find largest element in array of 10 bytes. [8 Marks]
b) Explain interrupt structure of 8086 microprocessor. [8 Marks]

14 MSE-II Examinations

[MSE-II papers for Microprocessor would go here if available]

15 SEE (Semester End Examinations)

15.1 CS3005-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

Q1. a) Explain 8086 architecture with Bus Interface Unit and Execution Unit. [8 Marks]
b) Explain memory segmentation in 8086. [8 Marks]
Q2. a) Explain different addressing modes of 8086 with examples. [8 Marks]
b) Explain instruction set of 8086. [8 Marks]
Q3. a) Write ALP for string operations. [8 Marks]
b) Write ALP for array processing. [8 Marks]
Q4. a) Write ALP for arithmetic operations. [8 Marks]
b) Explain interrupt handling in 8086. [8 Marks]
Q5. a) Explain 8255 PPI. [8 Marks]
b) Explain timer/counter operations. [8 Marks]

15.2 CS3005-1 SEE – May 2024

Semester End Examination, May 2024 Duration: 3 Hours — Max Marks: 100
CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

Q1. a) Explain microprocessor evolution. [8 Marks]
b) Explain 8086 pin diagram. [8 Marks]
Q2. a) Explain Bus Interface Unit. [8 Marks]
b) Explain Execution Unit. [8 Marks]
Q3. a) Explain memory organization. [8 Marks]
b) Explain stack operations. [8 Marks]
Q4. a) Explain procedure calls. [8 Marks]
b) Explain interrupts. [8 Marks]
Q5. a) Explain DMA. [8 Marks]
b) Explain serial communication. [8 Marks]

Part VI
INNOVATION AND DESIGN THINKING

16 MSE-I Examinations

16.1 ME1654-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025 Duration: 1 Hour — Max Marks: 15
ME1654-1 – INNOVATION AND DESIGN THINKING

Part A: Multiple Choice Questions (3 × 1 = 3 Marks)

1. Design Thinking approach primarily focuses on:
   (A) Following strict rules and guidelines
   (B) Understanding user needs and creating innovative solutions
   (C) Minimizing team collaboration
   (D) Avoiding experimentation

2. Which best describes "Empathy" in Design Thinking?
   (A) Finding faults in user behavior
   (B) Understanding users' feelings, thoughts, and motivations
   (C) Designing based on assumptions
   (D) Testing solutions without user involvement

3. The "Define" stage is primarily about:
   (A) Generating as many ideas as possible
   (B) Clearly articulating the problem based on insights from empathy
   (C) Building and testing prototypes
   (D) Observing users in their natural environment

Part B: Descriptive Questions (2 × 6 = 12 Marks)

UNIT – I

Q1. a) Explain the five stages of Design Thinking process with examples. [6 Marks]

b) What is the importance of empathy in Design Thinking? Explain methods to build empathy with users. [6 Marks]

UNIT – II

Q2. a) Describe the Ideation phase. Explain any two ideation techniques in detail. [6 Marks]

b) What is prototyping? Explain different types of prototypes used in Design Thinking. [6 Marks]

17 SEE (Semester End Examinations)

17.1 ME1654-1 SEE – May 2025

Semester End Examination, May 2025 Duration: 3 Hours — Max Marks: 100
ME1654-1 – INNOVATION AND DESIGN THINKING

Part B: Descriptive Questions

Q1. a) Explain Design Thinking process in detail. [8 Marks]
b) Explain Empathize stage methods. [8 Marks]
Q2. a) Explain Define stage techniques. [8 Marks]
b) Explain Ideate stage tools. [8 Marks]
Q3. a) Explain different Prototype types. [8 Marks]
b) Explain Test stage approaches. [8 Marks]
Q4. a) Explain innovation frameworks. [8 Marks]
b) Explain creativity techniques. [8 Marks]
Q5. a) Explain user research methods. [8 Marks]
b) Discuss design challenges. [8 Marks]

END OF QUESTION BANK

This comprehensive document contains all examination papers:

MSE-I: February 2025, February 2024, June 2023
MSE-II: April 2025, July 2023
SEE: May 2025, May 2024, July 2024, January 2023, August 2023
Supplementary: September 2022

Subjects: Linear Algebra, DBMS, DAA, SEPM, Microprocessor, Design Thinking

USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2025

MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS

(For AM, CB, CC, CS, IS, RI)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **In three dimensional Euclidean space inverse of (-2,-2,0) with respect to vector addition is**
   A) (2, 2, 2)
   B) (-1,-1,0)
   C) (-2,-2,0)
   D) (2, 2, 0)

2. **If a basis of a vector space has 3 elements, then dimension of the vector space is**
   A) 2
   B) 1
   C) 3
   D) 0

3. Let $T: R^3 \to R^4$ be the linear transformation defined by $T(x_1, x_2, x_3) = (x_1, 0, x_2, x_2)$. Then basis for null space of T is
   A) (1,0,0)
   B) (0,0,1)
   C) (0,1,0,0)
   D) (1,1,0,0)

4. Which of the following statement is not true if $T: U \to V$ is a linear transformation
   A) V is always the range of T
   B) Null space of T is a subspace of U
   C) Range space of T is a subspace of V
   D) $T(0)=0$

Part – B: Descriptive Answer Questions

*Note: Answer any **One** full question from **each Unit.***

			Marks	BT*	CO*	PO*
		Unit – I				
1.	a)	i) Define a subspace of a vector space V over a field F. ii) If G and H are subspaces of a vector space V over F then prove that $G \cap H$ is also a subspace of V.	4	L*1	3	1
	b)	Define (i) Row space (ii) Null space of a matrix. Check whether the vector $\begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$ is in the column space of the matrix $A = \begin{pmatrix} 1 & -3 & 4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix}$	4	L2	3	2
2.	a)	i) Define Linearly dependent and Linearly independent of a set of vectors. ii) Express $u=(1,-2,5)$ in R^3 as a linear combination of the vectors $u_1=(1,1,1)$ $u_2=(1,2,3)$ $u_3=(2,-1,1)$ (-6, 3, 2)	5	L1	3	2
	b)	Find the dimension of the subspace W of R^3 spanned by the vectors {(2,4,2), (1,-1,0),(1,2,1),(0,3,1)}. Extend the basis of W to a basis of R^3. [first 2 cols	3	L2	3	1
		Unit – II				
3.	a)	(i)State Rank-Nullity theorem. (ii) Let a linear transformation T: $R^3 \to R^3$ be defined by $T(x,y,z)=(x+2y-z, y+z, x+y-2z)$. Find the null space of T and range space of T. Also find dimension of the null space of T and dimension of range space of T.	5	L3	4	2

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b) Check whether a transformation T: $R^2 \to R^2$ defined by $T(x,y)=(x+y,x)$ is a linear transformation or not. 3 L2 4 1

4. a) Find the matrix of the linear transformation T: $R^3 \to R^2$ defined by $T(x,y,z)=(x-y+2z, 3x+y)$ with respect to bases $B=\{(1,1,1),(1,2,3),(1,0,0)\}$ and $B'=(1,1),(1,-1)$

$$\begin{bmatrix} 3 & -1 \\ 5 & 0 \\ 5 & 2 \end{bmatrix}$$

4 2

b) Let V and W be vector spaces over the field F. Let T and U be linear transformation from V into W. Then prove that the function (T+U) defined by (T+U)α=T(α)+U(α) is a linear transformation from V into W. Also prove , if c is any element of F, the function(cT) defined by (cT)(α)=c(Tα) is a linear transformation from V into W. 3 L2 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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T(1,1,1)

1-1+2 = 2
3+1 = 4

1,2,3
1-2+2(3) (5,5)
3+2

1,0,0
1-0+0 (1,3)

(2,3) = $a_{11}(1,1) + a_{21}(1,-1)$

$a_{11} + a_{21} = 2$ 5 1

$a_{11} - a_{21} = 4$ 5 3

$a_{11} + a_{11} + 1 = 2$

$6a_{11} = 6$ 3

$a_{21} = -1$

$$\begin{bmatrix} 3 & 5 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

a 5

2a

a_{11} a_{11}

a_{21}

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations-II, April 2025

CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(CS, CB)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following diagrams is used to model the vocabulary of a system?**
   A) Object Diagram
   B) Activity Diagram
   C) Class diagram
   D) Interaction Diagram
2. **Composition is a stronger form of which of the following?**
   A) Aggregation
   B) Inheritance
   C) Encapsulation
   D) Generalization
3. **Select the option that suits the Manifesto for Agile Software Development**
   A) Individuals and interactions
   B) Working software
   C) Customer collaboration
   D) All the above mentioned choices are correct
4. **What encapsulates both data and data manipulation functions?**
   A) Object
   B) Class
   C) Super Class
   D) Sub Class

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

			Marks	BT*	CO*	PO*
		Unit – I				
1.	a)	With a neat diagram explain Context model of MHC-PMS.	4	L2	3	2
	b)	Draw and explain the Sequence diagram for View Patient Information.	4	L2	3	2
2.	a)	Explain the Activity model of an Insulin's pump operation.	4	L2	3	2
	b)	Outline the Use case description for Report Weather.	4	L2	3	2
		Unit – II				
3.	a)	Explain various principles of Agile methods.	4	L2	3	2
	b)	Explain Test Driven Development Process with diagram.	4	L2	4	2
4.	a)	Explain any four Extreme Programming practices.	4	L2	3	2
	b)	Explain the Model of software testing process with diagram.	4	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CSE) Mid Semester Examinations - II, APRIL 2025

CS3005-1 – MICRPROCESSOR AND EMBEDDED SYSTEM

Duration: 1 Hour | Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following is NOT an advantage of modular programming?**
   A) Code reusability
   B) Easier debugging
   C) Increased execution time
   D) Better program organization
2. **Which interrupt is used to set video mode in 8086?**
   A) INT 10H, AH = 00H
   B) INT 16H, AH = 00H
   C) INT 21H, AH = 01H
   D) INT 13H
3. **Which register in ARM holds the flags indicating the status of the processor?**
   A) General-purpose register
   B) Program Counter (PC)
   C) Stack Pointer (SP)
   D) Program Status Register (PSR)
4. **What is the size of the general-purpose registers in ARM processors?**
   A) 8 bits
   B) 16 bits
   C) 32 bits
   D) 64 bits

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Compare and contrast RISC and CISC architectures	4	L2	3	1
b)	Write the procedure for reading a 2-digit hexa decimal number from the standard input device.	4	L3	3	1
2. a)	Discuss ARM design Philosophy.	4	L2	3	1
b)	Write an ALP to read an alphanumeric character and display its equivalent ASCII code at the center of the screen	4	L3	3	1
Unit – II					
3. a)	Write an embedded C program to control the speed of rotation of a stepper motor using potentiometer.	4	L3	4	1
b)	What is Arduino communication and explain the types of Arduino communication.	4	L2	4	1
4. a)	With a code snippet, explain pinMode(), digitalWrite(), and analogRead() functions.	4	L2	4	1
b)	Write an Embedded C program to scroll the word 'FIRE' from left to right on a 7-segment display.	4	L3	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

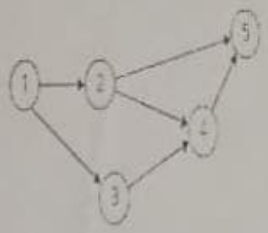
*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. **Which of the following is a necessary condition for a graph to have a topological sorting?**
   A) The graph must be a tree
   B) The graph must be a DAG
   C) The graph must be a complete graph
   D) The graph must be a bipartite graph
2. **What type of data structure is used in Depth-First Search (DFS)?**
   A) Queue
   B) Heap
   C) Priority queue
   D) Stack
3. **What is the time complexity of distribution counting sort?**
   A) $\Theta(n^2)$
   B) $O(n\log n)$
   C) $O(2^n)$
   D) $\Theta(n)$
4. **In dynamic programming, the solution to a problem is built up from:**
   A) A series of overlapping sub problems
   B) A single recursive call
   C) A greedy approach
   D) A divide and conquer method

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Outline Insertion sort algorithm and show the sorting of list E, X, A, M, P, L, E in alphabetical order using insertion sort method.	4	L3	3	2
b)	Apply heap sort algorithm to sort the list 90, 89, 56, 72, 67, 12, 21, 34, 45.	4	L3	3	2
2. a)	Define AVL tree and construct AVL tree for the data {2, 9, 3, 12, 15, 8, 11}	4	L3	3	2
b)	Define topological sorting. Illustrate the topological sorting using Source Removal method for the following graph.	4	L3	3	2



Unit – II					
3. a)	Write an algorithm to sort list of integers using Comparison Counting and compare its time complexity with the time complexity of selection sort and Bubble sort	4	L3	4	2
b)	Summarize the steps of Horspool string matching algorithm and search for the pattern TIME in the text A STITCH IN TIME SAVES NINE	4	L3	4	2

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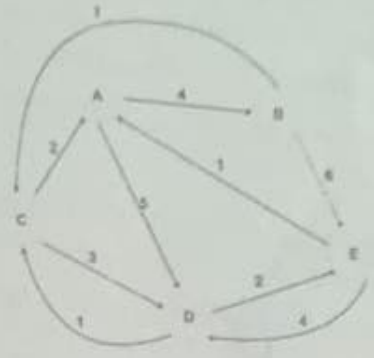
CS3004-1

MSE-II APRIL 2025

4. a) Apply All Pair Shortest path algorithm for the following graph.



b) Write an algorithm for Breadth-First Search (BFS). List any two facts that differentiate BFS from DFS. 4 L3 4 2

4 L3 3 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations - II, APRIL 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

CS, IS, AM, AD,CC

Duration: 1 Hour | Max. Marks: 15

Part – A: Multiple Choice Questions (1*3 = 3 marks)

*Note: Answer all **Three** questions in the **Answer Book**. Each question carries equal marks.*

1. **What is the result of the query SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE;?**
   A) Total number of employees
   B) Total number of salaries greater than a specified value
   C) Total number of unique salaries
   D) Average salary of employees
2. **What type of dependency does 2NF aim to eliminate?**
   A) Partial dependency
   B) Referential dependency
   C) Transitive dependency
   D) Candidate key dependency
3. **In a relation, if attribute B is functionally dependent on attribute A, which of the following is true?**
   A) Each value of B is associated with one value of A
   A → B
   B) Each value of A is associated with one value of B
   C) There is no dependency between A and B
   D) A and B must be the same attribute

Part – B: Descriptive Answer Questions (2*6 = 12 marks)

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Consider the following schema and write the SQL queries for the following: ***Suppliers*** (**sid**: integer, **sname**: string, **address**: string) ***Parts*** (**pid**: integer, **pname**: string, **color**: string) ***Catalog*** (**sid**: integer, **pid**: integer, **cost**: double) i. Find the *sids* of suppliers who supply some red or green part ii. Find the *sids* of suppliers who supply only red parts iii. Find the *pnames* of parts for which there is some supplier iv. Find the *sids* of suppliers who supply some red part or are located at 221 Packer Street. (**solution should be in Nested Query**)	6	L3	3	2
	a) Consider the following schema and write the SQL queries for the following: **Movie** (movie_id, title, release_year, genre, director_id, rating) **Director** (director_id, name, birth_year, nationality) **Actor** (actor_id, name, birth_year, nationality) **Movie_Actor**(movie_id, actor_id, role) **Review** (rev_id, user_id, movie_id, rev_text, rating, rev_date) **User** (user_id, username, email, age) i. Find the names of all actors who have acted in at least one movie released after 2015. ii. Find the directors who have directed more than 5 movies. iii. Get the average rating of all movies directed by 'Christopher Nolan'. iv. Find movies that have a higher rating than the average rating of movies released in the same genre. (**solution should be in Nested Query**)	6	L3	3	2

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CS2102-1 MSE-II APRIL 2025

Unit – II

L3	a)	Consider the two sets of functional dependencies $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$ Are they equivalent?	4	L3	4	2
	b)	What is Normalization? Explain the first Normal form with example.	2	L2	4	1
4	a)	Suppose a relational schema R (A B C D E F G H I J) and set of functional dependencies X: $\{ \{A,B\} \rightarrow C,\ A \rightarrow \{D,E\},\ B \rightarrow F,\ F \rightarrow \{G,H\},\ D \rightarrow \{I,J\} \}$ What is candidate key of R? Check out whether the relation is in 3NF or not? If not decompose it in 3NF.	4	L3	4	2
	b)	Explain any two Informal Design Guidelines for Relation Schemas.	2	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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4th sem MSE II